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Boron Embedded in Metal Iron Matrix as a Novel Anode Material of Excellent Performance

*Wujie Dong, Yantao Zhao, Xin Wang, Xiaotao Yuan, Kejun Bu, Chenlong Dong, Ruiqi Wang, and Fuqiang Huang**

Boron, the most ideal lithium-ion battery anode material, demonstrates highest theoretical capacity up to 12 395 mA h g^{-1} when forming Li_5B. Furthermore, it also exhibits promising features such as light weight, considerable reserves, low cost, and nontoxicity. However, boron-based materials are not in the hotspot list because Li_5B may only exist when B is in atomically isolated/dispersed form, while the aggregate material can barely be activated to store/release Li. At this time, an ingenious design is demonstrated to activate the inert B to a high specific capacity anode material by dispersing it in a Fe matrix. The above material can be obtained after an electrochemical activation of the precursors Fe_2B/Fe and B_2O_3/Fe. The latter harvests the admirable capacity, ultrahigh tap density of 2.12 g cm^{-3}, excellent cycling stability of 3180 mA h cm^{-3} at 0.1 A g^{-1} (1500 mA h g^{-1}) after 250 cycles, and superlative rate capability of 2650 mA h cm^{-3} at 0.5 A g^{-1}, 2544 mA h cm^{-3} at 1.0 A g^{-1}, and 1696 mA h cm^{-3} at 2.0 A g^{-1}. Highly conductive matrix promoted reversible Li storage of boron-based materials might open a new gate for advanced anode materials.

Boron, the lightest element that can alloy with Li apart from the hypertoxic beryllium, is one of the most ideal anode materials of lithium-ion batteries (LIBs). It has enormously high theoretical capacity of 2479 to 12 395 mA h g^{-1} for forming Li_yB (y = 1–5), and the further advantages of considerable reserves, low cost, and nontoxic nature. It seems that B is even better than the state-of-the-art candidates (e.g., Si: 4200 mA h g^{-1}; Al: 2234 mA h g^{-1}; Ga: 769 mA h g^{-1}; Ge: 1624 mA h g^{-1}; and Sn: 992 mA h g^{-1}),[1–4] which can alloy with Li and provide high specific capacity substitutes. However, B and its oxides are not in the hotspot list, because they can hardly be activated to store and release Li, Li-B alloys do exist and have been applied in thermal batteries.[5,6]

More or less efforts have been made to investigate the energy storage possibility of B_2O_3, B, or Li/B compounds as the anode materials of LIBs. Li and co-workers[7] used the pulsed laser deposition (PLD) method to deposit B_{50} thin film and harvested the capacity of only 43 mA h g^{-1} at room temperature, because of the poor lithium ion diffusion, but then greatly enhanced to 268 mA h g^{-1} at 85 °C. Liu et al.[8] investigated the electrochemical behavior of LiB compound anode materials. Even though high discharge capacity of 660 mA h g^{-1} can be attained, the charging capacity is only 274 mA h g^{-1}. In addition, the B_2O_3 thin film electrode, which was prepared by the PLD method, showed ultrahigh reversible capacities ranging from 1100 to 755 mA h g^{-1} all through the initial 100 cycles,[9] much better than the pure B_2O_3 powder sample (12 mA h g^{-1}).[10] However, the detailed analysis is little and the PLD method is obviously unsuitable for industrialization. Therefore, although some valuable attempts have been made, more efforts should be made to design new B-based anode materials with great performance.

Recently, we have developed high conductivity induced microstructure evolution of SnO_2-based materials, which ensured the excellent performance as an anode of LIBs.[11,12] It is established that the conductivity of anode materials exhibits great influence on their electrochemical reactions. In addition, it was reported that transition metal may improve the reversible lithium storage ability of oxides.[13,14] Combining the above thoughts, we suspect an ideal design to activate the B through dispersion of the B atoms in a conductive matrix. For example, iron, which is the most common and low-cost metal, can alloy with B but not Li, and thus may supply a stable and malleable matrix to make the above design come true. As shown in **Figure 1**, B atoms can fill in the lattices of Fe matrix, which can alloy with Li during the electrochemical reaction and form Li_yB clusters reversibly.

Herein, we first design and prepare a precursor Fe_2B/Fe, which can be activated to the ideal structure (Figure 1) progressively in the course of the electrochemical reaction. It

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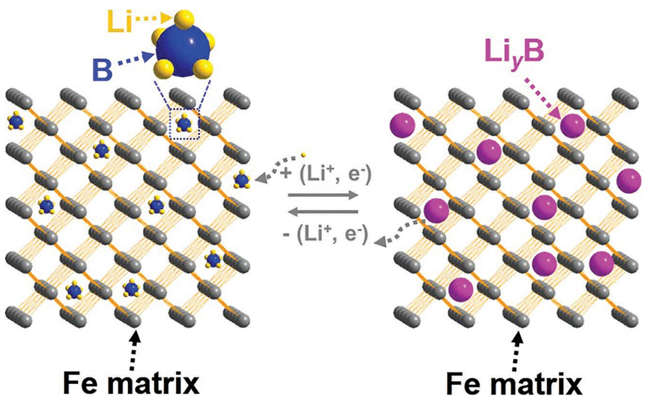

Figure 1. Schematic image of ideal design of B-based LIB anode material.

demonstrates that the inert B can be activated and it harvests high specific capacity (10 700 mA h g^{-1} after 1400 cycles, based on the mass of B) as the anode of LIBs. Seeing the low initial capacity (based on the total mass of Fe-B alloy) and long activation processing, we modify our design and fabricate the precursor of B_2O_3/Fe hybrid material, which can also be converted to a desired architecture (Figure 1) and possesses admirable initial capacity of 1696 mA h cm^{-3} (800 mA h g^{-1}) at 0.1 A g^{-1}, high tap density of 2.12 g cm^{-3}, excellent cycling stability of 3180 mA h cm^{-3} at 0.1 A g^{-1} (1500 mA h g^{-1}) after 250 cycles, and great rate capability of 2650 mA h cm^{-3} (1250 mA h g^{-1}) at 0.5 A g^{-1}, 2544 mA h cm^{-3} (1200 mA h g^{-1}) at 1.0 A g^{-1}, and 1696 mA h cm^{-3} at (800 mA h g^{-1}) 2.0 A g^{-1}. Highly conductive Fe matrix activates B_2O_3 to store/release Li reversibly at first, which is reduced to B during long-time lithiation/delithiation reaction. The Fe/FeO_x can gradually accept the O from Li_2O, which is the product of the B_2O_3/Li reaction, ensuring minor capacity loss of B_2O_3/B conversion. Later on, dispersed B in highly conductive Fe matrix lithiated/delithiated with Li, thus making the capacity increase repeatedly.

B and Fe can form the interstitial compound, where B is highly dispersed and infiltrates into Fe but retains the metallic conductivity. We expect this interstitial compound can provide a conductivity matrix and highly disperse B, thus enhancing its specific capacity. For the sake of verification, Fe-B (B mass ratio = 1–11 wt%) powder was prepared by a simple solid reaction method by mixing the Fe powder and B powder and vacuum sealed in a quartz tube. After reaction for 24 h at 800 °C in a muffle furnace, samples with 1–11 wt% B were obtained (**Figure 2**A), which involve two species of Fe and Fe_2B when B wt% < 9%, whereas pure Fe_2B when 9–11%. Fe (JCPDS No. 06-0696, space group *Im-3m* (229), $a = b = c = 2.8664$ Å, $\alpha = \beta = \gamma = 90°$) has a typical body-centered cubic (BCC) structure, and Fe_2B can be described as half of the body-centered Fe atoms are replaced by B, while the rest of the half is empty, consequently forming a 1D B chain in Fe matrix, as shown in Figure 2B. X-ray photoelectron spectroscopy (XPS) of Fe 2p and B 1s (Figure 2C) shows that some Fe on the surface is oxidized, but still remains Fe^0 (≈707 eV). Peak at ≈192 eV of B is the oxidized species[15] while the unoxidized B located at ≈188 eV, in accordance with the literatures.[16,17] The binding energy of B in Fe_2B is close to the pure B,[18,19] indicating the weak interaction between Fe and B in Fe_2B, which may lower the energy barrier in the subsequent lithiation. The obtained Fe-B alloy samples were then handled by high-energy ball milling in the argon atmosphere before applying them as active materials of LIBs. The particle size is reduced obviously, as shown in Figure S1A, C,D (Supporting Information).

Taking the pure B and Fe-B 1 wt% as an example, where pure B powder demonstrates slight and unstable capacity (Figure 2D, blue), it exhibits discharge capacity of 92 mA h g^{-1} initially and reduces to ≈6 mA h g^{-1} after 200 cycles at 0.1 A g^{-1} over a voltage window of 3.0–0.01 V in a 1 mol L^{-1} $LiPF_6$ electrolyte of 50:50 w/w mixture of dimethyl carbonate (DMC) and ethylene carbonate (EC), indicating the low activation of pure B in lithium ion storage. The poor performance of B could be ascribed to the huge Li^+ diffusion barrier[7] and its poor electronic conductivity, which permits only slight amount of B (on the interface of B particles and current collector) to participate in the electrochemical reaction. Figure 2D (red) shows the long-time cycling stability of Fe-B 1 wt% as anode materials. Even if, only 1 wt% of B, the initial reversible capacity is ≈30 mA h g^{-1} at 0.1 A g^{-1}(based on the total mass of Fe-B alloy), similar to the pure B (blue line). Moreover, while the capacity of B decreases rapidly, it increases continually for the Fe-B-1 wt% sample. After 1400 cycles, the specific capacity can reach up to ≈107 mA h g^{-1} and grow very slowly. The X-ray diffraction (XRD) pattern of the electrode and the XPS of B 1s after 1400 cycles are shown in Figure S1A,B (Supporting Information). The electrochemical performance of some other compositions (Fe-B alloy 7 wt% and 11 wt%) is shown in Figure S2A (Supporting Information). It is well known that the Fe cannot alloy with Li so that it has little capacity. If we employ the mass of B (1 wt%) to calculate the specific capacity, it is ≈3000 mA h g^{-1} at first and 10 700 mA h g^{-1} after 1400 cycles, which comes up to the high limitation of B's theoretical capacity (12 395 mA h g^{-1}). The trend of initial reversible capacity (based on the total mass of Fe-B alloy) is Fe-B 11 wt% (45 mA h g^{-1}) > Fe-B 7 wt% (40 mA h g^{-1}) > Fe-B 1 wt% (30 mA h g^{-1}), which means that the capacity increases on increasing the content of B in Fe-B alloy. This phenomenon is due to the increase in the content of active materials (Fe_2B). However, this increase is not linearly along with the increasing content of B, thus the trend of initial reversible capacity (based on the mass of B) is inversely Fe-B 11 wt% (409 mA h g^{-1}) < Fe-B 7 wt% (571 mA h g^{-1}) < Fe-B 1 wt% (3000 mA h g^{-1}). It means that most of B in Fe-B 11 wt% alloy is dead and useless at first because Li^+ can only infiltrate into the shallow surface (several nanometers) of Fe_2B and form Li_xB, but cannot reach inside of the material. Along with the long-term cycling, the big particles will crack because Li_xB induced huge volume expansion and exposed a new surface that Li^+ can infiltrate. Together with the further dispersion of B in Fe matrix, the capacity of Fe-B 11 wt% grows continually and faster than Fe-B alloy 7 wt% and Fe-B 1 wt% samples (the trend of growth rate based on the total mass during 800 cycles is Fe-B 11 wt% (8.26 mA h g^{-1} per 100 cycles) > Fe-B 7 wt% (5.83 mA h g^{-1} per 100 cycles) > Fe-B 1 wt% (3.34 mA h g^{-1} per 100 cycles), as shown in Figure S2A (Supporting Information). Similarly, since the most of B in Fe-B 11 wt% is dead and cannot store Li^+, the trend of growth rate by considering the active mass of B only is inversely, as shown in Figure S2B (Supporting Information). Capacities by considering the active mass of B only are calculated

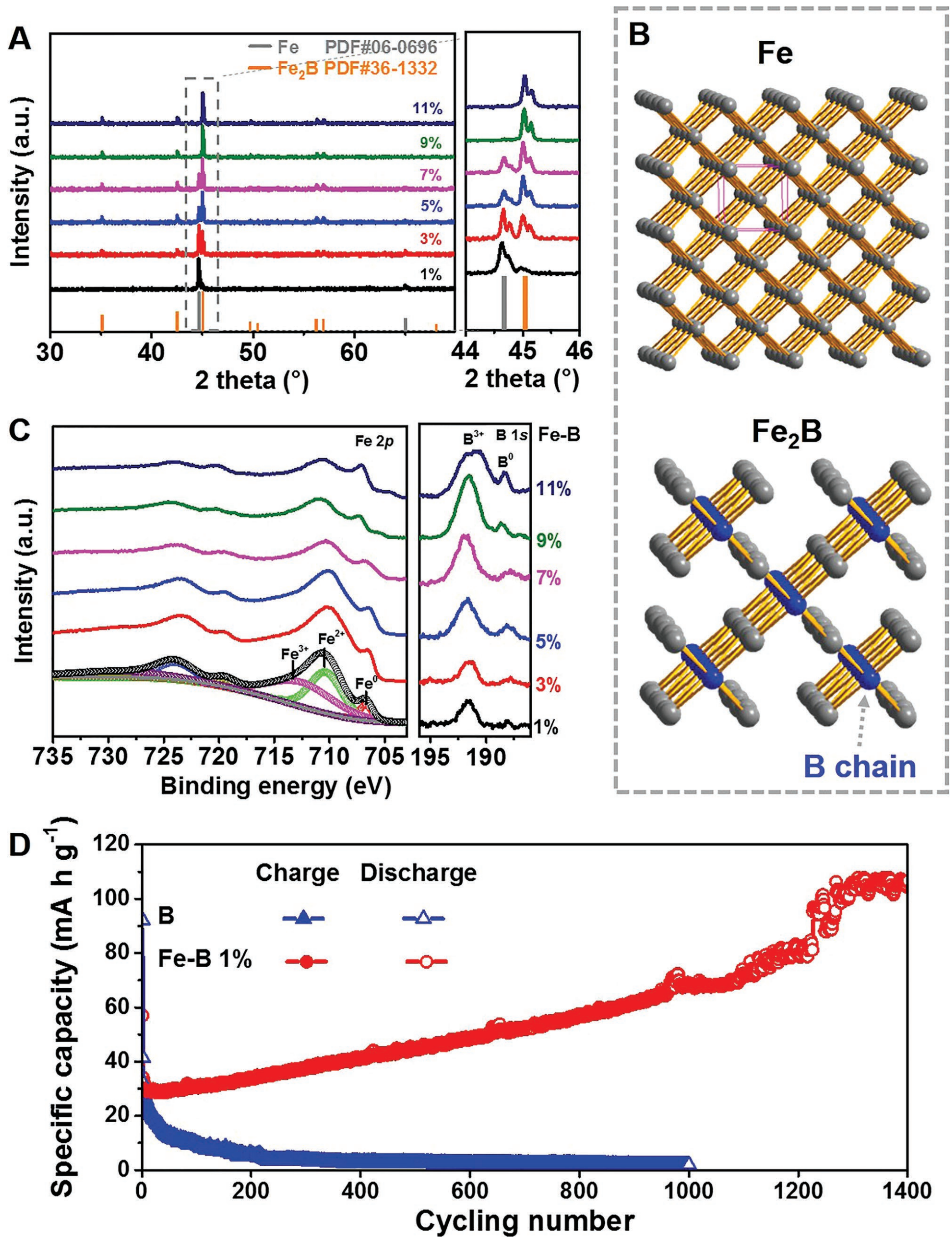

Figure 2. Physicochemical properties of Fe-B alloys and electrochemical performance of pure B and Fe-B-1 wt% electrodes. A) X-ray diffraction (XRD) patterns of Fe-B alloys (mass ratio of B from 1 wt% to 11 wt%), which are in accordance with that of Fe (JCPDS No. 06-0696, space group *Im-3m* (229), $a = b = c = 2.8664$ Å, $\alpha = \beta = \gamma = 90°$) and Fe_2B (JCPDS No. 36-1332, space group *Im/mcm* (140), $a = b = 5.1103$ Å, $c = 4.2494$ Å, $\alpha = \beta = \gamma = 90°$) shown at bottom. B) Crystal structure of Fe and Fe_2B. C) X-ray photoelectron spectroscopy (XPS) of Fe 2p and B 1s. D) Charge and discharge specific capacitance versus cycles at 0.1 A g^{-1}.

(Figure S2B, Supporting Information). For Fe-B alloy 7 wt% sample, it is 571 mA h g^{-1} at first and 1340 mA h g^{-1} after 850 cycles. For Fe-B alloy 11 wt% sample, it is 409 mA h g^{-1} at first and 1550 mA h g^{-1} after 1250 cycles.

Cyclic voltammetry (CV) curves of B, Fe-B-1 wt%, Fe-B-7 wt%, and Fe-B-11 wt% are shown in Figure S2C,F (Supporting Information), respectively, which are almost the same. There are some cathodic peaks at the first cycle of Fe-B-1 wt% sample. First of all, the biggest cathodic peak near 0 V belongs to the formation of Li_xB, while the small peak at ≈1.0 V is attributed to the breakdown of the boron lattice according to the literature.[7] As for the peak between 0.5 and 0.75 V, it may belong to the formation of solid electrolyte interface (SEI) since it disappears after the first cycle (Figure S2C,F, Supporting Information), which is well known. The small peaks at around 0.8, 1.4, and 2.4 V can be attributed to the lithiation reactions of B_2O_3 (come from the oxidation of surface B species, which is also confirmed by XPS, as shown in Figure 2C), which agree quite well with the plateaus of galvanostatic charging/discharging curves of B_2O_3 thin film.[9] The ref. [2] did not present the CV curves, but we found it at the master's thesis of the first author, which shows the peaks at the same positions. The detailed affiliations of each peak were not reported but they are obviously related to the stepwise lithiation reaction from B_2O_3 to $Li_xB_2O_3$ on the whole. The corresponding galvanostatic charging/discharging curves of 1st, 2nd, 10th, 100th, 1000th, and 1400th cycles (Figure S3B, Supporting Information) also approve the outstanding endurance of Fe-B-1 wt% electrode and show the major contribution of the capacity increment is in 0.01–0.5 V. The galvanostatic charging/discharging curves of the other three samples are shown in Figure S3 (Supporting Information). The large difference among B and Fe-B-1 wt% suggests that B can reversibly store and release Li. Compared with the pure B, Fe-B-1 wt% can provide a highly conductive matrix because of Fe, and the well-dispersed B can shorten the diffusion distance of Li^+. The B may further be dispersed constantly during the long-term cycle, as well as can be atomically dispersed to end with, so that the exceedingly high capacity of ≈10 700 mA h g^{-1} can be achieved, owing to the possibility of stable Li_5B[20] cluster formation under this condition.

Even though, the high specific capacity is achieved (based on the mass of B), the total performance of Fe-B-1 wt% is still far from the practical application. In addition, improving the contents of B in Fe-B alloys can enhance the capacity slightly but long activation processing is still needed (Figure S2A, Supporting Information). On the other hand, huge lithium storage is always accompanied by the huge volume expansion, for example, metal Sn electrode, which can be alleviated by adopting SnO_2 as the alternative, because when it lithiates, a Sn:Li_2O composite forms first before a Li-intercalated Sn alloy precipitates inside the Li_2O matrix, giving the hope that the matrix can help absorb the expansion stresses.[12] Inspired by this, B_2O_3 is assumed as the target material. However, as an electron insulator, B_2O_3 shows little electrochemical activity,[10] thus whether or not it can store Li is still a problem. Normally thermodynamics was employed to discuss the possibility of reversible redox reaction of metal oxides as the anode materials.[21] As of this viewpoint, if B_2O_3 can be activated to enforce the following analogous reaction

$$B_2O_3 + 6Li^+ + 6e^- \leftrightarrow 2B + 3Li_2O \quad (1)$$

The Gibbs free energy change $\Delta_r G_m^\circ = 3\Delta G\ (Li_2O) - \Delta G\ (B_2O_3) = -489.3$ kJ mol^{-1} (ΔH and ΔG of B_2O_3 are -1273.5 kJ mol^{-1} and -1194.3 kJ mol^{-1} at 298.15 K, respectively), indicating that Reaction (1) is spontaneous. Also, B/Li_2O mixture is desired to overcome an energy barrier of 489.3 kJ mol^{-1} to form B_2O_3 yet again, during the reverse reaction, much lower than that of 941.4 kJ mol^{-1} for Fe_2O_3 (the detailed calculation is provided in the Supporting Information) and 606.6 kJ mol^{-1} for SnO_2, where the former is a well-known reversible anode material and the latter was recently confirmed to be reversible under some conditions by ourselves[12] and Liu and co-workers.[21] Therefore, it is thermodynamically feasible to use B_2O_3 as an anode material of LIBs. The theoretical capacity of Reaction (1) is 2310 mA h g^{-1} if it is completely reversible, much higher than the other oxide anode materials (e.g., SnO_2: 1494 mA h g^{-1}; Fe_2O_3: 1005 mA h g^{-1}; Fe_3O_4: 928 mA h g^{-1}; and TiO_2: 335 mA h g^{-1}).[12,21–23] Likewise, the product B can store Li through the following reaction

$$B + yLi^+ + ye^- \leftrightarrow Li_yB \quad (2)$$

Here the value of y is quite controversial. Li-B alloys are generally recognized as a two-phase material: free metal Li fill in the fibrillar network framework of Li/B compound (from LiB to Li_5B).[5,6,8,20,24–27] Since the electrochemical property of this free metal Li is similar to the pure metal Li, only Li/B compounds are considered as the active materials. The theoretical capacity of B is between 2479 and 12 395 mA h g^{-1} for forming LiB to Li_5B (y = 1–5). Thus the total theoretical capacity range of B_2O_3 can reach up to 3080–6152 mA h g^{-1}, first level of all known anode candidates.

To active the electrochemical inert B_2O_3, we have designed a B_2O_3/Fe hybrid material, where the Fe matrix can provide a conductive network. As shown in **Figure 3**A (black line), the amorphous product was obtained primarily by an in situ reduction coprecipitation method, which possibly will comprise more or less $NaBH_4$, thus reducing the Fe (III) to metal Fe during the annealing at 500 °C in argon flow. XRD patterns (Figure 3A, red line) confirm the existence of Fe (JCPDS No. 06-0696, space group *Im-3m* (229), $a = b = c = 2.8664$ Å, $\alpha = \beta = \gamma = 90°$), while the other products look as if to be amorphous. Fe_2O_3, as the reference sample, can be obtained by a similar method without adding $NaBH_4$ (Figure 3A, blue line). Differently, Raman spectra can detect vibrations and rotational energy levels of molecules, which identifies the Fe_2O_3 species, both in unannealed and annealed samples, as illustrated in Figure 3B. Consistently, XPS of Fe 2p (Figure 3C) also confirm the coexistence of Fe $2p_{3/2}$ (III, ≈712 eV), Fe (II, ≈710 eV), and metal Fe (≈ 707 eV). The spectra of B (≈192 eV) suggest that it may exist as B_2O_3.[15] The peak positioned at ≈188 eV of unannealed Fe-B-O compound (UFBO) is the undecomposed $NaBH_4$, which disappears after annealing. However, since the atomic number of elemental B is too light and B_2O_3 is one of the most difficult species to crystallize, energy dispersive X-ray spectroscopy and transmission electron microscopy (TEM) cannot observe B. In order to assure the chemical composition of this hybrid material, inductively coupled plasma (ICP) emission spectrometry was accomplished and the molar ratio of Fe:B = 1:0.576. Thermo gravimetric analysis (TGA) curve (air flow; Figure S5, Supporting Information) of the annealed sample shows a little weight loss to 96.2% before 200 °C. To the best of our knowledge, the annealed sample contains some B_2O_3, which can adsorb water when exposed in air. The following weight increment up to 118.6% can be ascribed to the oxidation of Fe (0 or II) to Fe (III). This, combined with the results of ICP and TGA, finally determines the chemical composition of this hybrid material, which is $FeO_{0.324}\cdot 0.288B_2O_3$. The components can be considered as the metal Fe, iron oxide (FeO_x), and B_2O_3 (AFBO for short). Unlike the amorphous small nanoparticles of the unannealed one (UFBO; Figure S4A,B, Supporting Information), this AFBO sample shows the different morphology of sintered and larger particles (Figure 3E and Figure S4C, Supporting Information). Metal Fe and Fe_2O_3 can both be identified by interplanar crystal spacing and selected area electron

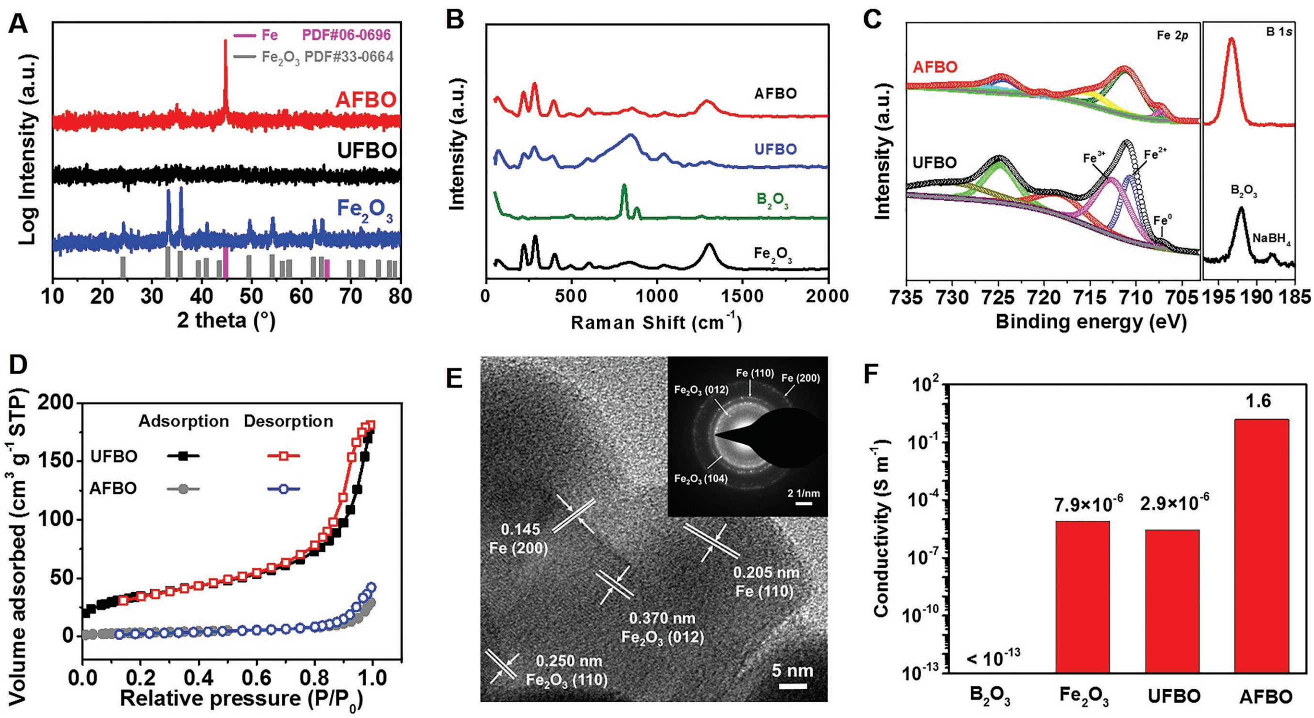

Figure 3. Composition of the prepared Fe-B-O hybrid material. A) X-ray diffraction (XRD) patterns of unannealed, annealed Fe-B-O hybrid (UFBO and AFBO) material, and annealed Fe_2O_3. The AFBO and annealed Fe_2O_3 show the phase of metal Fe (JCPDS No. 06-0696, space group *Im-3m* (229), $a = b = c = 2.8664$ Å, $\alpha = \beta = \gamma = 90°$). B) Raman spectra of some standard substance (Fe, Fe_2O_3, B, B_2O_3, and $NaBO_2$, both of them are analytical reagents) and the Fe-B-O hybrid material before and after annealing process. C) X-ray photoelectron spectroscopy (XPS) of Fe 2p and B 1s. D) N_2 adsorption–desorption isotherms of the UFBO and AFBO hybrid materials. E) TEM image of the annealed sample. Inset: Corresponding selected area electron diffraction (SAED) pattern. F) Electrical conductivity of pressed pellets.

diffraction (SAED) patterns. The annealing process also results in the huge change of specific surface area (unannealed sample: 126 $m^2\ g^{-1}$ and annealed sample: 13 $m^2\ g^{-1}$; Figure 3D) and tap density (unannealed sample: 1.36 g cm^{-3} and annealed sample: 2.12 g cm^{-3}). This big tap density has an advantage of volumetric energy density as the anode material when compared with some other nanosized materials or some other carbon-involved hybrid materials.[28–31] To measure the conductivity, *I–V* curves (Figure S6, Supporting Information) of the pressed pellets (≈0.5 g samples, $\Phi = 13$ mm, thickness ≈1 mm, 18 MPa) of B_2O_3, Fe_2O_3, UFBO, and AFBO samples were measured at room temperature. Since the conductivity of B_2O_3 is too low ($<10^{-13}$ S m^{-1}) and beyond the test limit of the instrument, we only obtained the data of self-prepared Fe_2O_3 (7.9×10^{-6} S m^{-1}), UFBO (2.9×10^{-6} S m^{-1}), and AFBO samples (1.6 S m^{-1}). As shown in Figure 3F, the apparent conductivity of AFBO is higher than that of the UFBO and Fe_2O_3 by about five to six orders of magnitude, thanks to the highly conductive Fe matrix. The extremely low conductivity of B_2O_3 confirms that it is the electron insulator that rarely has electrochemical activity.

Electrodes made of the above AFBO, UFBO, B_2O_3, and Fe_2O_3 exhibit totally different electrochemical performance than the anode materials of LIBs. **Figure** 4A shows the 1st, 2nd, 5th, and 250th CV curves of AFBO electrode over 3.0–0.01 V at a scanning rate of 0.3 mV s^{-1}. To compare, CV curves of B_2O_3, Fe_2O_3, and UFBO electrodes are also performed, as shown in Figure S7 (Supporting Information). In the first discharge cycle, the strong peak at 0.83 V can be attributed to the lithiation reaction of B_2O_3[9] and the relatively weak peak at 0.64 V is generally ascribed to the reduction of Fe^{3+} or Fe^{2+} to Fe^0 (Reaction (3))[32,33]

$$FeO_x + 2xLi^+ + 2xe^- \leftrightarrow Fe + xLi_2O \tag{3}$$

As measured before, $x = 0.324$, so that the theoretical capacity of FeO_x is 285 mA h g^{-1}. These two peaks shift to 1.47 and 0.91 V in the second and fifth cycles, because the electrode is activated in the first cycle and becomes easier to insert lithium. The weak peak near 0 V is similar to Figure S2 (Supporting Information), contributed by Reaction (2), whereas the wide weak at 1.58 V in the first charge cycle is the reverse process of Reaction (4). The primary peak at 2.01 V is due to the oxidation of B. Interestingly, the CV curve becomes quite different after 250 cycles and the area is much bigger than the initial five cycles. While the peak at 1.47 V is reduced, the peak that belongs to Reaction (3) is enhanced, indicating that O may be transferred from B_2O_3 to FeO_x. More remarkable variation occurs near 0 V, where a much intensive peak appears and contributes to the main capacity. Galvanostatic charging/discharging curves of AFBO, UFBO, Fe_2O_3, and B_2O_3 show the one-to-one corresponding plateaus, as shown in Figure S8 (Supporting Information).

To identify the above suspects of each peak about boron, the most effective way is monitoring the binding energy of B by

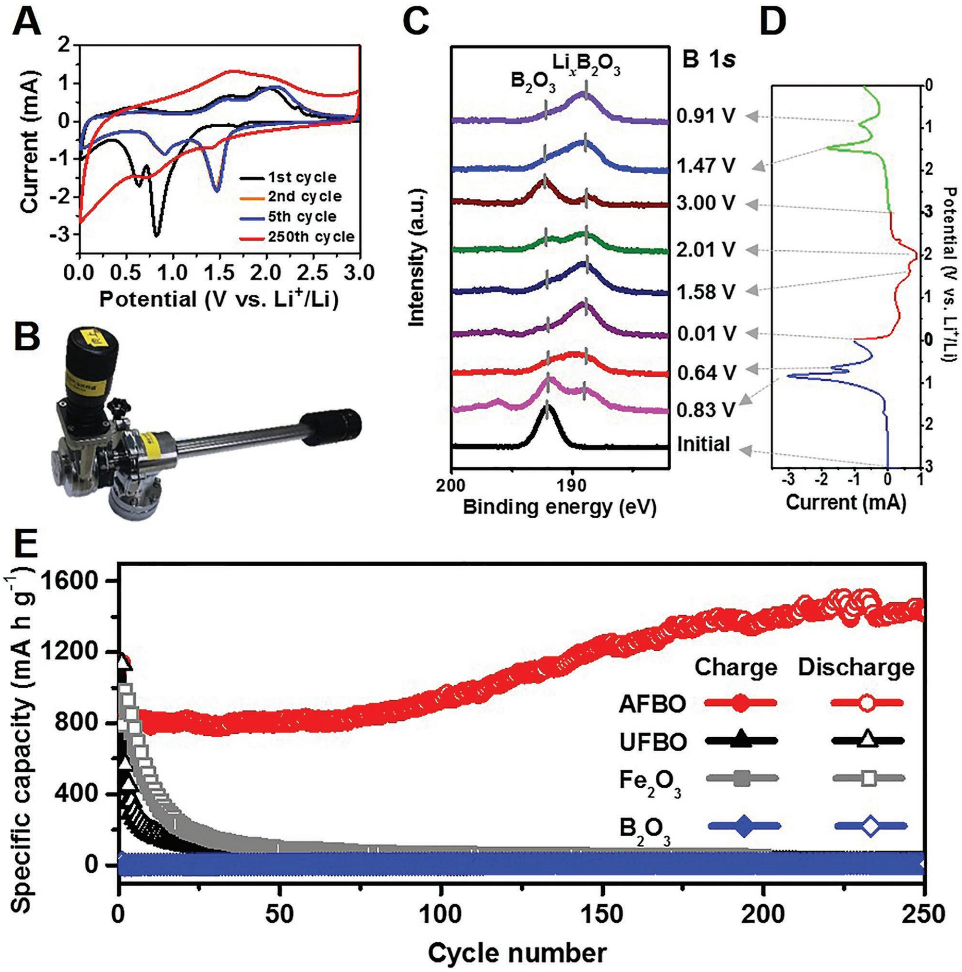

Figure 4. Electrochemical performance of AFBO electrode. A) The 1st, 2nd, 5th, and 250th CV curves at a scanning rate of 0.3 mV s^{-1}. B) Image of inert gas (Ar) injection rod, which can load the samples in the inert gas (Ar) glove box and then transfer them into the high vacuum ($\approx 10^{-9}$ Pa) test chamber of XPS without exposure into air. C) B 1s XPS spectrum of the AFBO electrodes at different cutoff potential, which are corresponding with D) the unfolded CV curves of 1st discharge (blue)/charge (red) curves and 2nd discharge (green) curves. E) Galvanostatic charge/discharge specific capacity versus cycles at 0.1 A g^{-1} of AFBO, UFBO, and Fe_2O_3 electrodes.

XPS, which can show the valence state evolution along with the charge/discharge processing. Although in situ XPS is a perfect choice to do this experiment, the current technique does not allow us to do so, since XPS is only suitable for the shallow surface analysis. As a less-than-ideal alternative, we used inert gas (Ar) injection rod (Figure 4B), which can load the samples in the inert gas (Ar) glove box and then transfer them into the high vacuum ($\approx 10^{-9}$ Pa) test chamber of XPS without exposure into air. A lot of coin cells were first assembled, run the CV curves, and then stopped at different cutoff potential. These cells were open in an argon glove box, where AFBO-loaded electrode disks were then handled to the XPS samples and loaded on the inert gas (Ar) injection rod. Figure 4C,D shows the B 1s XPS spectrum of the AFBO electrodes at different cutoff potential and the unfolded CV curves of 1st discharge (blue)/charge (red) curves and 2nd discharge (green) curves, respectively. The binding energy of B 1s in initial AFBO is ≈192 eV, corresponding with B_2O_3. After discharge to 0.83 V, a shoulder peak appears at ≈189 eV whose relative intensity increases along with the reduction of potential to 0.64 and 0.01 V. On the contrary, the peak at ≈192 eV becomes weak from 0.83 to 0.01 V, indicating that the valence state of B decreases from +3 in B_2O_3 to a lower value ($Li_xB_2O_3$, B, or Li_yB). After discharge to 0.01 V and then recharge to 1.58 V, the B 1s XPS spectrum barely changes, suggesting this peak may belong to the oxidation of Fe. Once it comes to 2.01 V, the peak at ≈192 eV increases while the peak at ≈189 eV decreases, showing that a part of boron is oxidized at this potential. When it loops back to 3.00 V, a majority of B comes back to ≈192 eV (B_2O_3), but still retains a little low valence state species, which may be B^0. At the 2nd discharge to 1.47 V, the peak at ≈192 eV becomes weak and the peak at ≈189 eV is considerably enhanced, confirming the lithiation reaction of B_2O_3.

The Nyquist plots, spanning over 0.01 to 10^5 Hz (5 mV excitation), of the AFBO electrode before and after initial five CV cycles are shown in Figure S9B (Supporting Information), which are fitted by Zview (Figure S9A in the Supporting Information shows Randles equivalent circuit for these cells). The superlative electronic conductivity of AFBO (Figure 3F) confers a much better AC impedance to the electrochemical cells when compared with B_2O_3, Fe_2O_3, and UFBO electrodes (Figure S9C–E, Supporting Information). The charge transfer resistance of initial AFBO electrode is ≈200 Ω, which is reduced to ≈20 Ω, indicating the significant activation effect of the first cycle on weakening the charge transfer barrier. This lower barrier guarantees that the lithiation reactions become stress free (at higher potential) and homogeneous. The long-term cycling stabilities of B_2O_3, Fe_2O_3, UFBO, and AFBO electrodes at 0.1 A g^{-1} are exhibited in Figure 4E, where B_2O_3 shows little specific capacity below 20 mA h g^{-1} because it is an insulator. Fe_2O_3, however, has a high reversible capacity of ≈1000 mA h g^{-1}, but fades rapidly because of huge volume change during the lithiation/delithiation reaction. UFBO has the specific capacity of >1100 mA h g^{-1} at first discharge cycle, but decreases to <100 mA h g^{-1} within 30 cycles. Differently, the reversible capacity of AFBO electrode is ≈800 mA h g^{-1} initially, which rises gradually and reaches up to ≈1500 mA h g^{-1} after 200 cycles and then remains stable. To estimate by its tap density (2.12 g cm^{-3}), the specific volumetric capacity is as high as 3180 mA h cm^{-3}.

The AFBO electrode also performed well under higher current densities of 0.2, 0.5, 1.0, 2.0, 5.0, and 10 A g^{-1}, with the specific capacity recorded as 850, 810, 750, 680, 550, and 430 mA h g^{-1}, respectively, as shown in Figure S10 (Supporting Information). After reverting to 0.1 A g^{-1}, the capacity is also recovered and then increases to ≈1500 mA h g^{-1} within a further 100 cycles. Since the capacity retention ratios are excellent at 0.5, 1.0, and 2.0 A g^{-1}, long-term cycling stability at these high current densities was also carried out. As per the statement, the first ten cycles are tested at 0.1 A g^{-1} because the small current density may help form better SEI and make the electrode more homogeneous. As shown in **Figure 5**A, the specific capacities under 0.5 and 1.0 A g^{-1} are growing gradually, similar to 0.1 A g^{-1} (Figure 4E), which can finally come up to ≈1250 (2650 mA h cm^{-3}) and ≈1200 mA h g^{-1} (2544 mA h cm^{-3}), respectively. As for the 2.0 A g^{-1}, the specific capacity can hold steady at ≈800 mA h g^{-1} (1696 mA h cm^{-3}) after the fast increment at first. To be honest, as shown in Figure 5B, the specific gravimetric capacity of AFBO (≈1500 mA h g^{-1}) is in a high level of the reported works,[12,14,15,28–31,33–41] but is still lower than the groundbreaking works based on Si from Cui and co-workers[34,35] and Li alloy/graphene anode materials (≈1600 mA h g $^{-1}$).[42] However, considering the low tap density of these nanomaterials or carbon-involved hybrid materials (listed in Figure 5, / means the tap density was not reported), the specific volumetric capacities of these materials are much lower than our AFBO (annealed $Fe/FeO_x/B_2O_3$) hybrid materials in this work, which is 3180, 2650, 2544, and 1696 mA h cm^{-3} at 0.1, 0.5, 1.0, and 2.0 A g^{-1} after 250 cycles, respectively, as shown in Figure 5C. Just as known, the specific volumetric capacity of metal Li is 2061 mA h cm^{-3} because of its low density (0.534 g cm^{-3}).[43] These extraordinary and excellent performances of AFBO make this work the first level of LIB anode researches.

Furthermore, to be clear, the theoretical capacity of AFBO ($FeO_{0.324}\cdot 0.288B_2O_3$) is based on three parts: FeO_x (Reaction (3), 285 mA h g^{-1}), B_2O_3 (Reaction (1), 2310 mA h g^{-1}), and B (Reaction (2), between 2479 and 12 395 mA h g^{-1} for forming LiB to Li_5B). Accordingly, the total theoretical specific capacity of AFBO can be calculated after considering their weight ratio respectively, which are between 1121 and 2444 mA h g^{-1}. On the grounds of the CV curves (Figure 4A) and galvanostatic charging/discharging curves (Figure S8A, Supporting Information), the major contribution of the initial capacity is attributed to Reaction (1) (B_2O_3), and the second one is from Reaction (3) (FeO_x) while Reaction (2) (B/Li_yB) is the least. The reason is that the B^0 is difficult to form by Reaction (1), instead, $Li_xB_2O_3$ ($x \le 6$) is the main product during lithiation reaction,[9] and returns to B_2O_3 during the reverse reaction, as suggested by the XPS spectrum of B 1s (Figure 4C). Figure 4A shows the good reversibility of the peak at 1.47 V, which can be identified as the formation of $Li_xB_2O_3$. However, a little amount of B° can be formed in each cycle (Figure 4C) and cumulative during long-time cycling (Figure S11, Supporting Information), which results in the decrease of peak at 1.47 V (discharge cycle) and the remarkable increase of peak near 0 V after 250 cycles, as shown in Figure 4A. Fe, however, can react with Li_2O and accept O by reverse Reaction (3), where the O is originally from B_2O_3. As shown in Figure 4A, redox couple peaks at ≈0.9 V (discharge cycle) and ≈1.5 V (charge cycle) are obviously enhanced after 250 cycles, indicating that there are more O in FeO_x. The gradually increased B^0 seems to be dispersed by Fe^0, just like the Fe-B interstitial compound discussed before, making the whole electrode amorphous (Figure S12, Supporting Information).

The intrinsic microstructural evolution mechanism during the electrochemical lithiation/delithiation of Fe-B alloy and AFBO electrodes are difficult to observe directly for now. After combining the above experimental and analytical data, here we bring out our assumptions about the microstructure evolution of B-based anode materials: For the Fe-B alloy electrode, Fe and Fe_2B species co-exist. Fe (JCPDS No. 06-0696, space group *Im-3m* (229), $a = b = c = 2.8664$ Å, $\alpha = \beta = \gamma = 90°$) has a typical BCC structure and cannot alloy with Li. The Fe_2B can be described as half of the body-centered Fe atoms are replaced by B while the another half is empty, thus forming a 1D B chain in Fe matrix, as shown in **Figure 6**A. Since B can alloy with Li, Li_yB species forms after the discharge processing (Figure 6A, reaction I), where y is close to 1, thus harvesting the specific capacity of ≈3000 mA h g^{-1} based on the mass of B (Figure 2C). As is well known, anode materials with alloying mechanism are usually accompanied by significant volume change during the lithiation/delithiation reaction; the same happens for B. The huge volume change may result in great stress, break the B chain and the lattices of Fe_2B, generate a little isolated B atoms or clusters, and force them to be the interstitial atoms into the lattice of Fe, just like $Fe_{1.8}B_{0.2}$ (ICSD No. 613897) reported before.[44] The isolated B atom may combine with 5 Li, forming high stable Li_5B cluster, as suggested by Tai and Nguyen.[20] However, it is a long-term process but not accomplished in an action, thus making the specific capacity increase progressively until up to ≈10 700 mA h g^{-1} (Figure 6, reaction III). During the above reactions, the electrode materials will crack because of the volume change induced significant stress, which can be detected by XRD (Figure S1, Supporting Information). As for the AFBO electrode, the highly conductive Fe matrix induces the high activity of B_2O_3, which can store numerous Li by forming $Li_xB_2O_3$ (Figure 6B, reaction I). This stated reaction

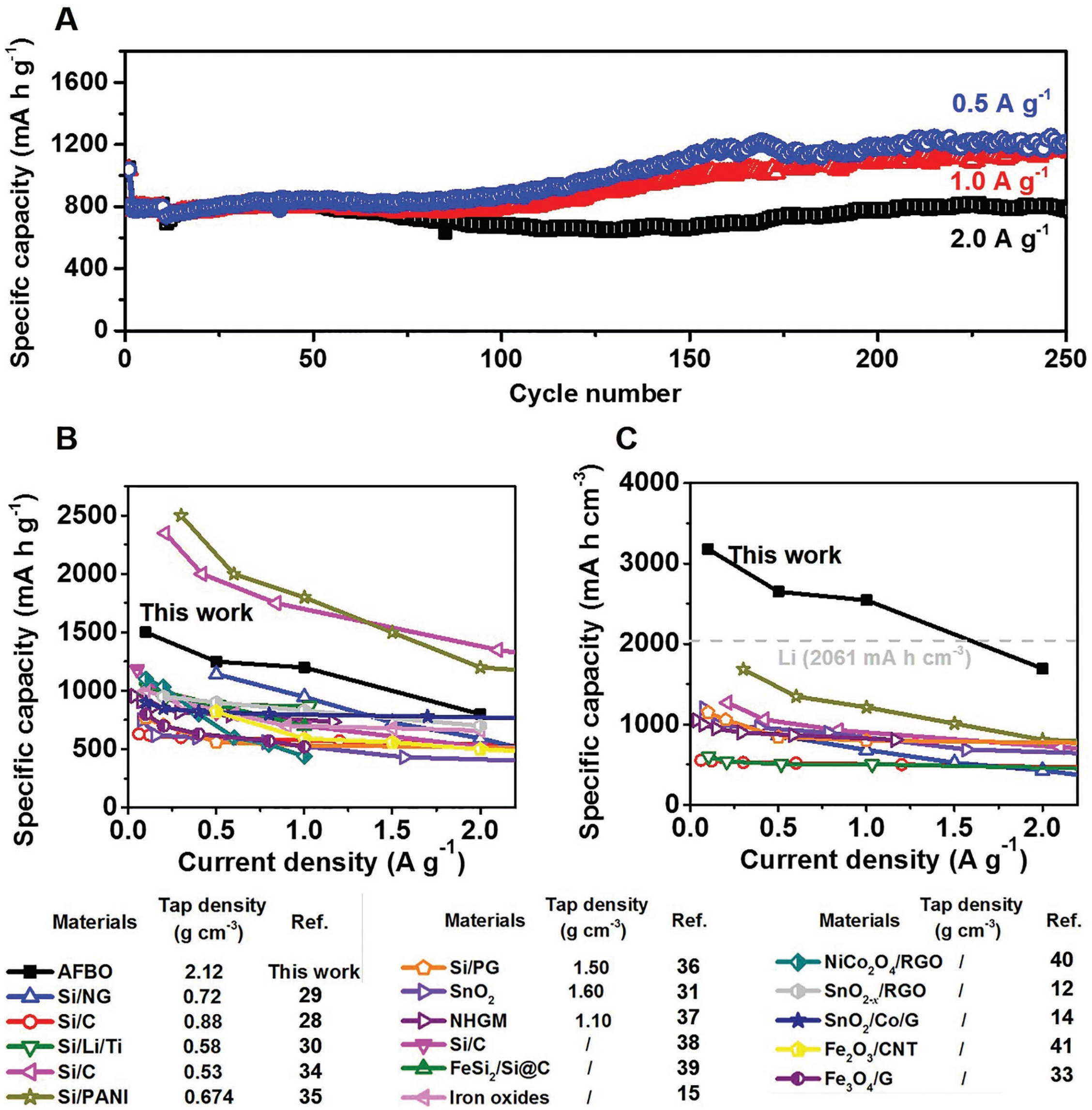

Figure 5. A) Long-term cycling stability of AFBO electrodes at the current density of 0.5, 1.0, and 2.0 A g^{-1}. Comparison of the specific B) gravimetric and C) volumetric rate capability among the reported materials[12,14,15,28–31,33–41] and our AFBO (annealed Fe/FeO$_x$/B$_2$O$_3$) hybrid materials in this work.

contributes the major capacity of this hybrid material. It is believable that a part of B$_2$O$_3$ is reduced to B^0, which can disperse into Fe and accumulate during the long-term cycling. As described in Fe-B alloy, these B species can further combine more Li, confirmed by CV peak near 0 V (Figure 4A). Therefore, the gradually increased capacity of AFBO electrode comes from the growing B/Li$_y$B reaction (Figure 6, reaction III). The insertion of B together with the further lithiation/delithiation breaks the lattices of Fe, thus making the whole electrode amorphous (Figures S11 and S12, Supporting Information). To conclude, iron has the following functions: (1) conductive matrix; (2) a lithium-inert and malleable host of B/Li$_y$B, which can disperse B, ease the volume change, and suppress the aggregation during lithiation/delithiation; (3) in the Fe/B$_2$O$_3$ system, metal Fe is the acceptor of O (derived from B$_2$O$_3$), and prevents the capacitance loss of the electrode.

In summary, although B and B$_2$O$_3$ have the extremely high theoretical capacity as the anode materials of LIBs, they are out of the research hotspot because of their electrochemical inertness. Here we first prove that B (1%) can be activated when alloying with Fe and show the specific capacity of 10 700 mA h g^{-1} if only counting the mass of B. Toward the practical application, a B$_2$O$_3$/Fe hybrid material is developed, which possesses admirable initial capacity of 1696 mA h cm^{-3} (800 mA h g^{-1}) at 0.1 A g^{-1}, high tap density of 2.12 g cm^{-3}, excellent cycling stability of 3180 mA h cm^{-3} at 0.1 A g^{-1} (1500 mA h g^{-1}) after 250 cycles, and great rate capability of 2650 mA h cm^{-3} (1250 mA h g^{-1}) at 0.5 A g^{-1}, 2544 mA h cm^{-3} (1200 mA h g^{-1}) at 1.0 A g^{-1}, and 1696 mA h cm^{-3} at (800 mA h g^{-1}) 2.0 A g^{-1}, respectively. The volumic capacity of this hybrid material is even beyond the popular Si-based anode materials. We suspects that the highly conductive Fe matrix activates B$_2$O$_3$ to storage/release Li reversibly at

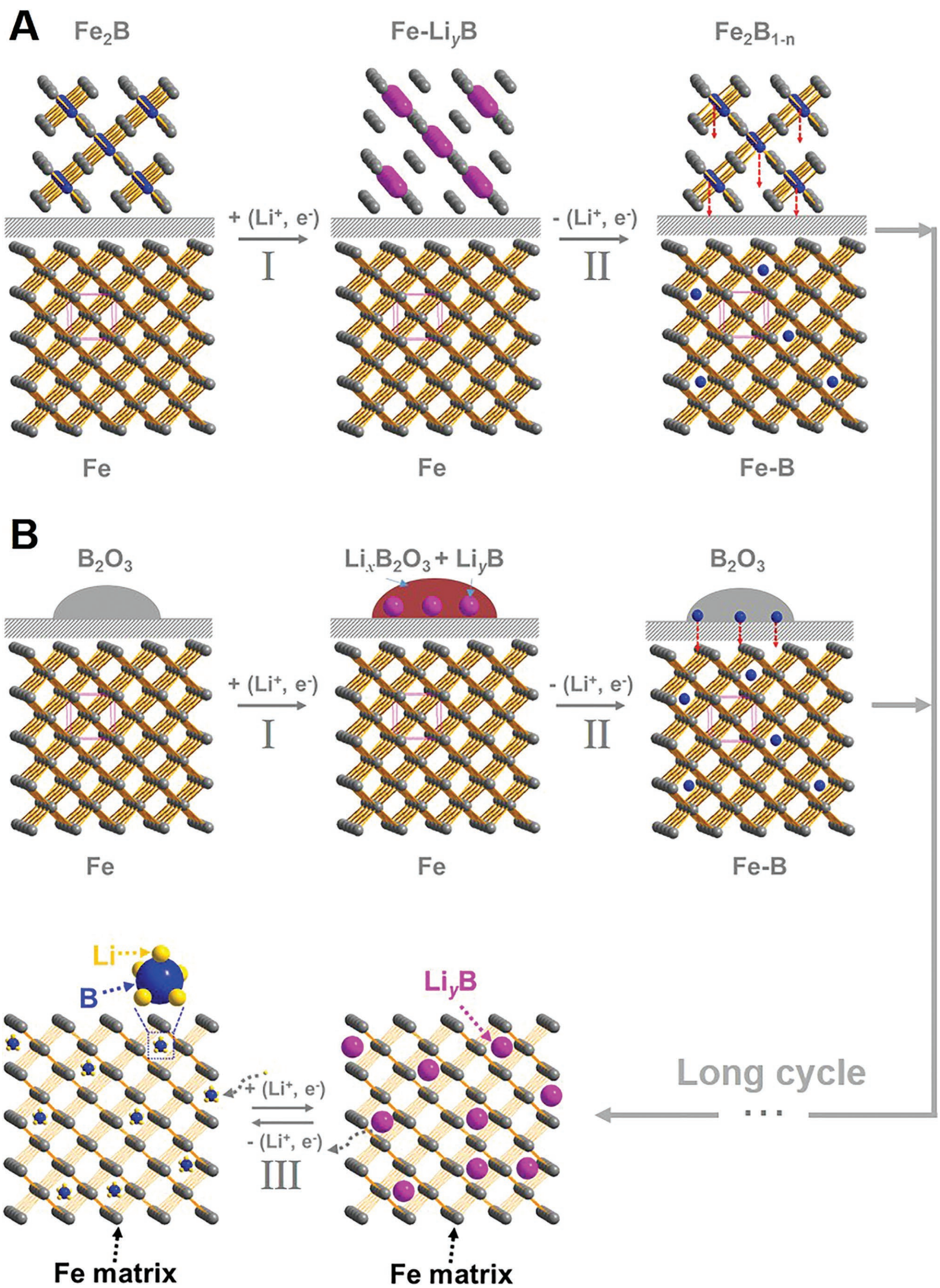

Figure 6. Schematic microstructure evolves during the charging/discharging process of A) Fe-B 1% and B) AFBO electrodes.

first, which is reduced to B and dispersed into Fe lattices during long-time lithiation/delithiation reaction. This atomically dispersed B in highly conductive matrix can lithiate/delithiate with Li, thus making the capacity increase continually. To be honest, further research about B-based anode materials is still needed when facing the practical application since smooth and steady capacity are more favored. How to develop the ability of B-based anode at initial cycles and keep it stable during the long-term cycling are still great challenges. In addition, the deep mechanism needs to be explored and understood. Nevertheless, highly conductive matrix promoted reversible Li storage of boron and its oxide may open a new gate for advanced anode materials, and it may inspire wider explorations on some other anode materials system (e.g., atomic or cluster-level dispersed carbon or some other element may bring out surprising specific capacity).

Experimental Section

All chemicals used in this work were analytical pure without any further purification and commercially available. Deionized water used was house generated. (Resistivity: 0.860 MΩ cm at 25 °C according to Rex DDSJ-308F Conductivity Meter from INESA Scientific Instrument Co.)

Synthesis of Fe-B Alloys: Fe-B alloys with different B content were prepared by a simple solid state reaction. Briefly, for the Fe-B 1%

sample, 0.01 g B and 0.99 g Fe powder were mixed by hand milling for half an hour. The mixed powder was then added into quartz tube and flame-sealed under vacuum (1×10^{-3} mbar). The sealed quartz tube was heated at the rate of 1 °C min^{-1} until up to 800 °C in a programmable furnace. After reacting at this temperature for 24 h, it cooled down naturally until room temperature. The particle size of the obtained powder is quite big so as to handle with high energy ball milling for 2 h in Ar atmosphere before acting as the active materials of LIBs. Other Fe-B alloy samples with different B content were prepared in the same way but only adjusting the ratio of Fe and B.

Synthesis of B_2O_3 /Fe Hybrid Material: B_2O_3/ Fe (AFBO) hybrid material was prepared by annealing the UFBO, which was synthesized by a simple hydrolysis co-precipitation method. Specifically, 2.70 g $FeCl_3{\cdot}6H_2O$ (0.01 mol) was dissolved in 150 mL deionized water (DI). 6 mL $NH_3{\cdot}H_2O$ solution (28%) was added into the $FeCl_3$ solution under stirring. After stirring overnight, the above solution was mixed with 20 mL $NaBH_4$ aqueous solution, which dissolved 1.89 g $NaBH_4$ (0.05 mol). The $NaBH_4$ solution was freshly prepared every time before use in an ice bath to prevent decomposition. The color of the solution became black after adding $NaBH_4$ solution and generated a lot of gas. After further stirring for 5 h, the precipitates were separated from the solution by centrifugation, washed by DI water, and freeze dried for 24 h. The obtained sample was named UFBO, and after annealing at 500 °C in tube furnace (Ar flow, 300 mL min^{-1}) for 4 h, AFBO was obtained. To prepare Fe_2O_3, the method is almost the same but not adding $NaBH_4$.

Powder XRD patterns were collected using a Bruker D2 X-ray diffractometer with Cu Kα radiation (k = 1.5418 Å) at a scanning rate of 2° min^{-1} in the 2θ range of 10°– 80°. Nitrogen adsorption–desorption isotherms at 77 K were measured by an Accelerated Surface Area and Porosimetry System (ASAP2020, Micrometer) using vacuum-degassed samples (120 °C for at least 4 h). These isotherms were used to calculate the specific surface area by the Brunauer–Emmett–Teller method. XPS (Axis Ultra of Kratos Analytical Ltd. Al Kα radiation, $h\nu$ = 1486.7 eV) was calibrated in situ using the 284.8 eV peak of delocalized sp^2-hybridized carbon that appeared on every sample due to atmospheric contamination. This procedure was repeated at the beginning of every XPS data analysis. The microstructures were characterized by TEM (JEM2010-HR). The chemical composition of the hybrid material was confirmed through inductively coupled plasma-atomic emission spectrometer (Prodigy 7 of Leeman Ltd.) to measure the atomic ratio of Fe and B. Both the UFBO and AFBO sample were measured by TGA at a heating rate of 10 °C min^{-1} in the temperature range of 25–1000 °C under air atmosphere using a Thermal Analysis Q600SDT. Raman spectra were collected on a Thermal Fisher Micro Raman imaging spectrometer (DXRxi) using a laser with an excitation wavelength of 532 nm.

An *N*-methyl pyrrolidinone slurry consisting of 80 wt% active material (Fe-B alloys, UFBO, and so on), 10 wt% acetylene black, and 10 wt% polyvinylidene fluoride was uniformly coated to a copper foil. The coated foil was dried at 60 °C for 12 h in vacuum oven and then cut into disk electrodes of 14 mm in diameter. After that, the electrodes were further vacuum-dried overnight at 60 °C. AFBO electrode disk was obtained by annealing the UFBO electrode disk at 500 °C in tube furnace (Ar flow, 300 mL min^{-1}) for 4 h, just the same as the powder sample. Lithium foil (China Energy Lithium Co., Ltd.) was used as the counter and reference electrode, and a glass fiber mat (Whatman) was used as separator. The electrolyte used was 1 mol L^{-1} $LiPF_6$ in a 50:50 w/w mixture of DMC and EC. Coin cells were assembled in a recirculating argon glove box, which reserved the moisture and oxygen contents below 1 ppm. Electrochemical measurements were carried out on a LAND-CT2001C test system. The cycle lifetime of cells was characterized at different rates within a voltage window of 3.00–0.01 V. The rate capability was evaluated by varying the discharge–charge rate from 0.1 to 10 A g^{-1}. Electrochemical impedance spectroscopy (EIS), room-temperature *I–V* characteristics, and CV of the assembled coin cells were carried out using an automated electrochemical workstation (CHI660E). The EIS was measured between 0.01 and 10^5 Hz with an excitation voltage of 0.005 V at various potentials. The room-temperature *I–V* characteristics were accomplished in the voltage range of –0.3 to +0.3 V at a scanning rate of 1 mV s^{-1}. The CV curves were obtained at a scanning rate of 0.3 mV s^{-1} over the potential from 0.01 to 3.00 V versus Li^+/Li.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

boron oxide, Fe-B alloy, high conductivity, Li-B alloy, lithium ion batteries (LIBs)

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